

UAV-RT field review

Prototype report: what was built, the methods, and how it did against known tag positions

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In one paragraph. After a flight the operator wants to know where to walk next. This prototype reads the pulse log TagTracker already writes, filters it by tag, time, altitude and a map box, interpolates the signal strength over the flown area, and reports the position where the received signal was strongest, in decimal degrees, together with any other lobe worth knowing about, a bearing with a confidence, and contour lines. On the flights where the true tag position is known, the strongest signal landed 22 m to 41 m from the tag on three well-sampled Cumbria flights and 23 m and 22 m on the final flights of the two Ponui kiwi localisations, in every case on the downhill side of the tag, as the field paper predicts. All numbers in this document were produced by `python/validate_fieldreview.py`; MATLAB was never run.

Contents

1	Why	2
2	What was built	2
2.1	The control set	2
2.2	What the operator gets	2
3	Methods	4
3.1	Local frame and interpolation	4
3.2	Smoothing	4
3.3	Grid spacing from the data	4
3.4	Strongest signal, and other lobes	5
3.5	Bearing	6
3.6	Contours	6
3.7	Terrain check, validation only	6
4	Results	7
4.1	Ground truth	7
4.2	Against truth	7
4.3	Downhill?	10
4.4	Sensitivity to grid spacing and smoothing	10
4.5	Contours at small scale	11
4.6	Two lobes	11
5	Answers to the specification's questions	13
6	For the TagTracker delivery	13
7	Caveats and open items	13
A	Command line	13
B	Regenerating this document	13

1 Why

The field workflow that motivates this is in `docs/FIELD_REVIEW.md` in the hub repository. Between flights the operator lands, copies the pulse CSV off the Herelink to a laptop, opens MATLAB, eyeballs the strongest signal, and reads coordinates off the screen while cross-referencing a phone map. Each flight's result decides where the next one goes, and the laptop round trip sits in the middle of that loop. The two New Zealand kiwi localisations reported in Mohammadi, Shafer and Flikkema (2026) took four and three flights each in exactly this loop.

The data is already on the Herelink. This prototype settles what the on-device analysis should compute before anything is written into TagTracker. Its deliverable is settled behaviour, not a user interface: the four questions the specification asks are answered in Section 5.

2 What was built

Everything lives in `uavrt_postflight/python/` and is plain Python with numpy and matplotlib, the same dependencies as the bench app.

File	Purpose
<code>analysis.py</code>	Shared primitives, new: <code>select_pulses</code> , <code>grid_spacing</code> , <code>peak_prominence</code> , <code>find_peaks</code> , <code>edge_distance</code> , <code>contour_polylines</code>
<code>fieldreview.py</code>	The entry point. Pipeline, text report, figure, command line
<code>dem.py</code>	Terrain sampling for validation only; not part of the field feature
<code>validate_fieldreview.py</code>	Runs every case with known truth and writes this document's figures and tables
<code>test_fieldreview.py</code>	51 checks on synthetic surfaces with known answers plus a real log
<code>kmzwrite.py</code>	Now takes its contour lines from the shared routine

The bench app (`pulseplotter.py`) was not touched, and neither was TagTracker. The log reader, the geodesy module, and the existing `build_grid` and `estimate_bearing` are used exactly as they were.

2.1 The control set

Deliberately small, as the specification asks.

- **Tag.** Required in practice; the default is the most common tag in the log.
- **Time window.** Opens on the flight proper: the takeoff climb and the landing descent are trimmed by the median-altitude rule the bench app already uses. The operator can widen or narrow it.
- **Altitude window.** For the case the specification calls out, where the path recrosses the launch point and near-field pulses contaminate the surface.
- **Spatial box.** The rectangle an operator would draw on the map. Implemented in the filter and on the command line; no drawing gesture in this prototype.
- **Confirmed only.** On by default. Every log on disk contains only confirmed pulses, and the system reference says the CSV never receives unconfirmed ones, so this filter cannot be exercised on real data.

Grid spacing, smoothing window and the lobe thresholds are also on the command line, for the configuration menu, not the main screen.

2.2 What the operator gets

Figure 1 is the figure for Cumbria tag 42 and the box below is the text report for the same run. Positions are decimal degrees to five places, about 1 m.

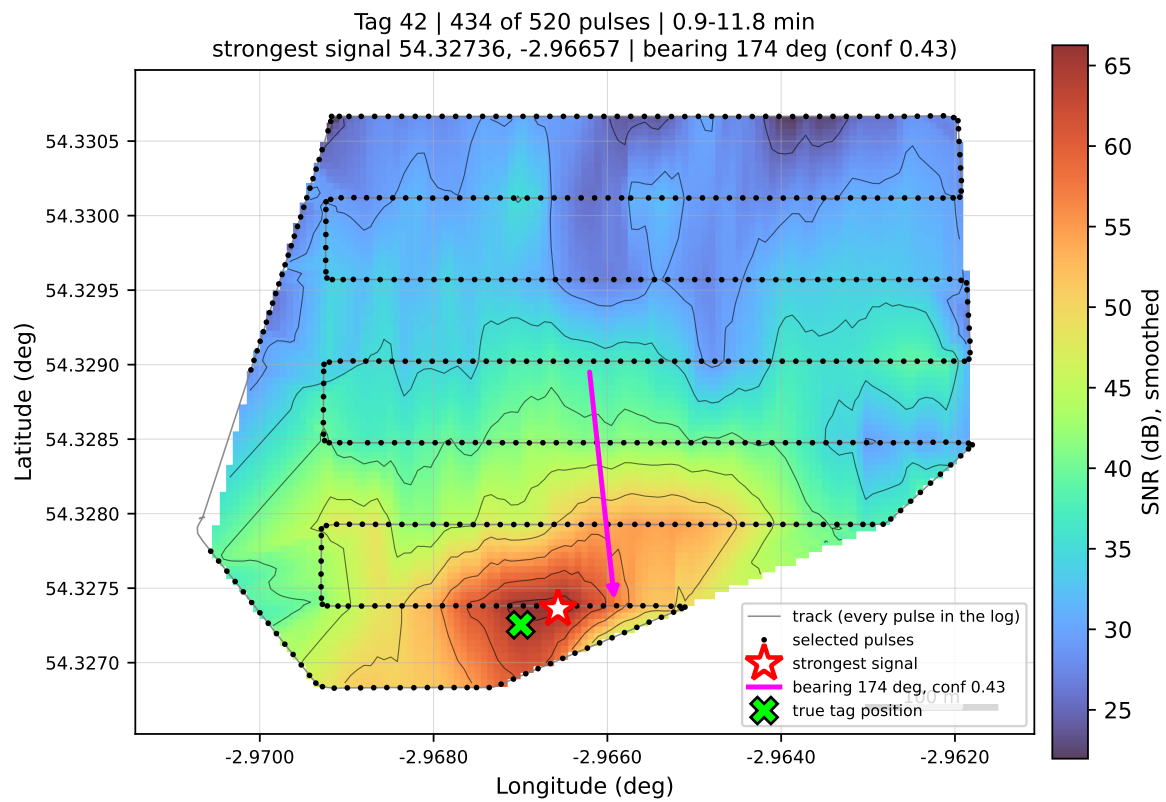


Figure 1: Cumbria, 21 November 2025, 11:08 log, tag 42. The raster is the interpolated smoothed SNR, transparent outside the flown area; thin lines are contours; the grey line is the track through every pulse in the log and the black dots the pulses that passed the filters. The star is the strongest signal, the cross the true tag position, the arrow the bearing from the centre of the selection. The strongest signal is 30 m east-north-east of the tag.

```

Pulse-2025-11-21-11-08-41-149.csv: 1111 pulses, 13.3 min, tags 40 (490), 41 (65), 42 (520), 43 (36)
Tag 42: 434 of 520 pulses selected
  confirmed only; time 0.9-11.8 min (flight proper, takeoff and landing trimmed); altitude any; box none
Grid 5.0 m (half the 10.0 m median pulse spacing; extent 570 x 428 m), 114 x 86 cells; smoothing 6 pulses
Lobes reported when within 10 dB of the strongest and at least 2 dB above the saddle joining them to higher
ground, at most 3
Peak SNR 66.3 dB, surface median 35.2 dB, contrast 31.0 dB

STRONGEST SIGNAL 54.32736, -2.96657 SNR 66.3 dB
  This is where the signal peaked, not where the tag is. Expect the tag uphill of it, typically 20-30 m
  away (Mohammadi 2026).

BEARING 174 deg (S) from 54.32895, -2.96620, the centre of the selection; confidence 0.43, spread +-75
deg

Truth 54.32725, -2.96700: strongest signal is 30 m away, to the ENE (67 deg) of the tag
  bearing to truth from the selection centre 195 deg; estimate off by -21 deg

```

Three things the report always does. It labels the position *strongest signal* and says on every run that this is not the tag position and why. It flags a *weak* surface, one whose peak is less than 10 dB above the median of the surface, as unreliable. And it says when the peak sits on the edge of the flown area, because then the surface was still rising when the survey ran out and the useful advice is to fly further that way.

3 Methods

3.1 Local frame and interpolation

Positions are converted to a local east–north frame anchored on the first pulse of the log, using the WGS84 meridional and normal radii of curvature at the reference latitude:

$$x_E = R_N \cos \phi_0 (\lambda - \lambda_0), \quad y_N = R_M (\phi - \phi_0), \quad (1)$$

which agrees with a rigorous ECEF transform to under 5 cm over a kilometre. The smoothed SNR is interpolated onto a regular grid by linear interpolation over the Delaunay triangulation of the pulse positions (`build_grid`, unchanged), and is undefined outside their convex hull.

One property of that surface matters for everything that follows: **the maximum of a piecewise-linear interpolant is always at a vertex**. The strongest-signal position is therefore always the position of a pulse, after smoothing, and never a point between flight lines. Its cross-track resolution is the transect spacing. This is also why the grid spacing barely moves the reported position (Section 4.4): the grid only quantises where that vertex is sampled.

3.2 Smoothing

SNR is smoothed in flight order with the bench app’s centred moving mean over $k = 6$ pulses,

$$\bar{v}_i = \frac{1}{|W_i|} \sum_{j \in W_i} v_j, \quad W_i = \{j : i - \lfloor k/2 \rfloor \leq j \leq i + k - \lfloor k/2 \rfloor - 1\}, \quad (2)$$

windows shrinking at the ends and NaNs omitted, matching MATLAB’s `movmean`. This is smoothing along the track, not across it.

3.3 Grid spacing from the data

A fixed spacing in metres looks wrong on one of a 60 m lawnmower and a kilometre survey. Two things bound a sensible spacing. The surface carries no detail finer than the pulse spacing, so half the median distance d between consecutive pulses is the natural starting point; it is also the PI’s rule of thumb (about 10 m at 15 mph and 1.5 s per pulse). And the map is drawn at a fixed size

whatever was flown, so fewer than about 60 cells across the longer side L looks blocky and more than about 300 is invisible and slow:

$$\Delta = \text{clip}\left(\frac{d}{2}, \frac{L}{300}, \frac{L}{60}\right), \quad \Delta \geq 1 \text{ m.} \quad (3)$$

Consecutive pulses at exactly the same position, which the 2023 logs produce because the position feed was slower than the pulse rate, are left out of the median. The existing cell cap in `build_grid` still applies on top. Table 1 shows what the rule chose on every dataset on disk.

Dataset	Pulses	Extent (m)	Step (m)	Grid (m)	Bound	Cells	Time (s)
Cumbria 09:35 (largest survey)	265	1060 × 1140	14.9	7.4	step/2	22022	0.02
Cumbria 11:08	434	570 × 428	10.0	5.0	step/2	9804	0.03
Ponui case 2 flight 3	328	223 × 316	4.9	2.5	step/2	11739	0.03
Ponui case 1 flight 4	363	184 × 159	4.7	2.4	step/2	5451	0.01
Raymond Park scan 2025-01-13	524	262 × 296	5.0	2.5	step/2	12614	0.04
Engineering lawn lawnmower 2025-01-03	166	67 × 65	3.7	1.1	extent/60	3599	0.01
Raymond Park monopole 2024-10-11	655	91 × 469	2.3	1.6	extent/300	17759	0.04
NAVHDA 2023-08-18 (1 km, headerless)	1012	1003 × 1087	11.1	5.5	step/2	35854	0.04

Table 1: Grid spacing chosen by equation (3) across the datasets on disk, the bound that decided it, and the time for the whole review on the development laptop. Every survey landed on half the pulse spacing; only the two smallest flights and the 2023 log with duplicated positions fell back to an extent bound.

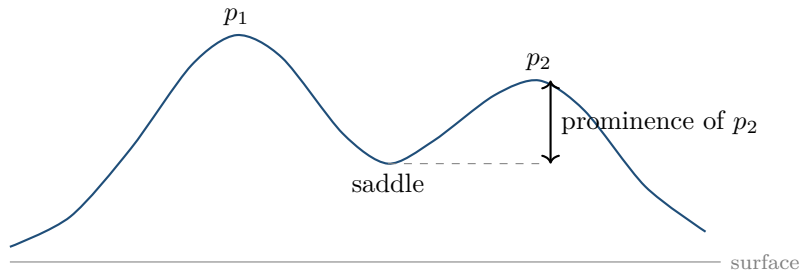
3.4 Strongest signal, and other lobes

The strongest signal is the highest cell of the surface. The question the specification asks is what to do when the surface has two lobes: two transmitters on one channel, a reflection, or a transit leg. Reporting only the highest cell hides the other; reporting every local maximum drowns the operator, because a real surface has hundreds of them (163 on the surface of Figure 8).

What separates a second lobe from a wrinkle on the first is its **topographic prominence**: how far the surface has to drop from that maximum before it can climb to something taller. For a local maximum p and any higher point q ,

$$\text{prom}(p) = z(p) - \max_{\gamma: p \rightarrow q, z(q) > z(p)} \min_{c \in \gamma} z(c), \quad (4)$$

the height of p above the highest saddle that connects it to higher ground. The global maximum has no higher ground; its prominence is taken as its height above the lowest cell.



It is computed as zero-dimensional persistence. Cells are visited from the highest down. A cell with no visited 8-neighbour starts a new component and is a local maximum; otherwise it joins the components of its visited neighbours, and when that joins two components the one with the lower peak dies at the current level, its prominence being its peak minus that level. Union-find with path compression makes this $O(N \log N)$ for the sort and near-linear after it; NaN cells are never visited, so nothing crosses the edge of the flown area. On the grids here it takes a few tens of milliseconds.

A local maximum is then **reported as a lobe** when

$$z_{\max} - z(p) \leq W = 10 \text{ dB} \quad \text{and} \quad \text{prom}(p) \geq P = 2 \text{ dB}, \quad (5)$$

at most three in all, ranked by height. Each carries how far below the strongest it is, the distance and compass direction from the strongest, and its prominence. An optional minimum separation in metres is implemented but off. The thresholds were chosen from the sweep in Table 4: at the 3 dB window first proposed, no well-sampled flight ever reports a second lobe, including the two-transmitter surface where the second transmitter is 6.5 dB down; at 10 dB and 2 dB that surface reports both sites, every clean bullseye still reports one lobe, and the flights that spray lobes are the ones already flagged weak.

Two further flags come from the same surface. **Weak:** the peak is less than 10 dB above the median of the surface, so the surface is essentially flat and its maximum is noise; every flight in the validation where the tag was only heard near the noise floor was caught by this. **Edge:** the peak is within two cells of a NaN cell or the array border, i.e. of the hull of the flown area.

3.5 Bearing

Unchanged from the bench app and restated here because it is on the screen. With $\mathbf{g} = \nabla z$ the gradient of the surface, which points uphill toward the tag, and cells outside the hull excluded,

$$\mathbf{R} = \frac{\sum_i |\mathbf{g}_i| \hat{\mathbf{g}}_i}{\sum_i |\mathbf{g}_i|} = \frac{\sum_i \mathbf{g}_i}{\sum_i |\mathbf{g}_i|}, \quad \theta = \text{atan2}(R_x, R_y), \quad c = |\mathbf{R}|, \quad \sigma = \sqrt{-2 \ln c}. \quad (6)$$

θ is a compass bearing, c the confidence in $[0, 1]$ (1 when every cell agrees) and σ the circular standard deviation. The magnitude weight cancels the normalisation, so the direction is that of the mean gradient; what the circular form adds is c . The bearing is anchored at the centre of the selected pulses.

The validation adds one thing worth knowing about it. When the tag is *inside* the flown area the surface is a bullseye, the gradients point every way, and the confidence collapses: 0.13 on the final Ponui flight, with the bearing 115° wrong on the flight before. The bearing is the right tool when the tag is beyond the survey and the surface is a slope; the peak is the right tool when it is inside. The confidence tells the two apart.

3.6 Contours

Levels are the same nice values matplotlib would pick for an integer level count, kept strictly inside the data range. The lines are extracted with `contourpy`, the library matplotlib itself uses, as plain coordinate arrays, first in local metres and then in lat/lon, with no figure involved, so a map renderer, the KML writer and the plot all draw the same lines. The KMZ writer now uses this routine.

3.7 Terrain check, validation only

The field paper’s explanation for the offset is propagation geometry: the monopole’s toroidal pattern and the terrain put the strongest received signal slightly downhill of the tag. To test that, `dem.py` samples two elevation models that need no login: the Environment Agency’s 1 m LiDAR terrain model for Cumbria, fetched by lat/lon box from their WCS service, and Copernicus GLO-30 (30 m) for Ponui, a surface model that includes tree canopy. At the true tag position a plane $z \approx a + b e + c n$ is fitted to the cells within 25 m (EA) or 60 m (Copernicus); the downhill direction is $\text{atan2}(-b, -c)$. The report gives $\Delta z = z(\text{peak}) - z(\text{tag})$ and the component of the tag-to-peak offset along the downhill direction. Δz is the more direct of the two: the terrain is not a plane over 30 m, and on the Cumbria knoll the two disagree in sign.

4 Results

4.1 Ground truth

Site	Position (lat, lon)	Source	Note
Cumbria tags 40, 41	54.326601, -2.9684094	the PI's Tag42_Above_60m_PROCESS.m, beside the logs	one site, two tags
Cumbria tags 42, 43	54.327254, -2.966997	same	one site, two tags
Ponui case 1 kiwi	-36.887851, 175.184943	PI's field notebook, 4 Sep 2026	under a bush; may have moved during the day
Ponui case 2 kiwi	-36.886383, 175.178570	PI's field notebook, 4 Sep 2026	in a burrow

The paper's case 1 is four ten-minute flights on one day; case 2 is three. The Ponui logs match that on 20 March (tag 12, four flights of 8–14 minutes, the first with its strongest detections 1.5 min after takeoff, as the paper describes) and 23 March (tag 24, three flights with median altitudes stepping 36, 25, 3 m). The case 1 position was first supplied with a digit wrong; the flights themselves pointed at a position 29 m from the one the notebook then gave, which is a small validation of its own.

4.2 Against truth

Case	Pulses	Peak SNR	Contrast	Offset	Dir.	Δz	Downhill	Bearing err.	Lobes
Cumbria 11:08, tag 42	434	66 dB	31 dB	30 m	ENE	-6 m	-4 m	-21°	1
Cumbria 11:08, tag 43 (weak)	13	11 dB	5 dB	481 m	NE	-32 m	-285 m	+166°	1
Cumbria 11:08, tag 40	429	49 dB	25 dB	85 m	N	+2 m	-29 m	-37°	2
Cumbria 11:08, tag 41 (weak)	61	11 dB	9 dB	562 m	NE	-33 m	+160 m	+130°	3
Cumbria 09:35, tag 42 (edge)	265	56 dB	14 dB	124 m	ESE	-23 m	+64 m	+71°	2
Cumbria 09:35, tag 40 (weak)	268	35 dB	8 dB	190 m	E	-15 m	+174 m	+30°	3
Cumbria 13:05, tag 42	451	63 dB	27 dB	22 m	NE	-3 m	-7 m	-15°	1
Cumbria 13:05, tag 40	449	48 dB	23 dB	85 m	N	+3 m	-34 m	-17°	2
Cumbria 14:12, tag 42	451	61 dB	21 dB	41 m	ENE	-7 m	-7 m	-12°	1
Cumbria 14:12, tag 40	452	48 dB	19 dB	127 m	NE	+2 m	+60 m	-17°	1
Ponui case 1, flight 1 (edge)	179	34 dB	21 dB	195 m	NE	+18 m	+184 m	+40°	2
Ponui case 1, flight 2 (weak)	314	38 dB	9 dB	158 m	NNE	+18 m	+156 m	+0°	3
Ponui case 1, flight 3	346	45 dB	16 dB	42 m	NNE	-19 m	+42 m	+86°	2
Ponui case 1, flight 4	363	51 dB	15 dB	23 m	NNE	-11 m	+23 m	+64°	2
Ponui case 2, flight 1 (weak)	86	20 dB	7 dB	119 m	ESE	+1 m	+64 m	-28°	2
Ponui case 2, flight 2	293	32 dB	17 dB	36 m	SE	-3 m	+29 m	-115°	1
Ponui case 2, flight 3	328	32 dB	18 dB	22 m	E	-2 m	+4 m	+25°	1

Table 2: Every case with a known tag position, at the defaults. Offset is from the tag to the strongest signal, with its compass direction; Δz is the elevation of the peak minus that of the tag; downhill is the component of the offset along the local downhill direction at the tag, positive when the peak is on the downhill side; bearing error is the estimate minus the true bearing from the centre of the selection to the tag. *Weak* and *edge* are the report's own flags.

Cumbria tag 42, the well-sampled case. On the three 60 m–120 m surveys of the same box the strongest signal is 30 m, 22 m and 41 m from the tag, all to the north-east, all 3 m to 7 m lower in elevation. The tag sits on a small knoll (Figure 5); the peak is on its slope. The 09:35 flight, flown higher and over four times the area, gives 124 m with the edge flag set: a coarse first pass, which is what it was.

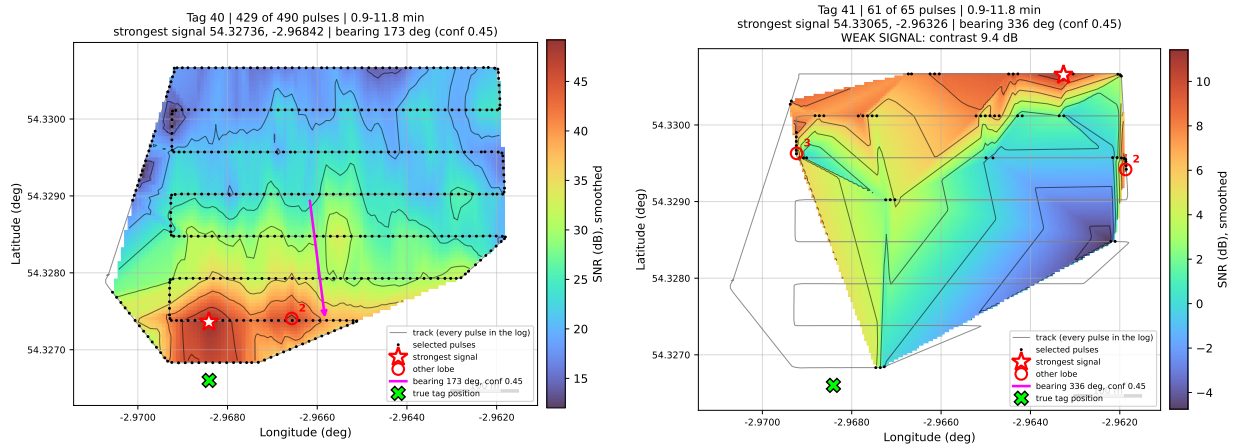
Cumbria tag 40, the tag beyond the survey. The tag is 85 m south of the southernmost transect, and the peak lands on that transect directly north of it on all three flights (Figure 2a).

This is the cross-track limit of the method: the surface cannot peak where nothing flew. The one leg that did pass within 25 m of the tag, the outbound leg at low altitude, heard it *more weakly* than the transect 85 m away; that is the overhead null of the vertical monopole, and it is the same physics as the downhill offset. Without smoothing the single strongest pulse is on that near leg, 33 m from the tag.

Cumbria tags 41 and 43, the weak case. Both were only heard near the noise floor, 11 dB peak and 5 dB to 9 dB contrast. The positions are 480 m to 560 m wrong and the report says so: weak, at the edge, with the bearing 130° off. Figure 2b is what an operator would see.

Ponui case 2. The three flights converge 119, 36, 22 m (Figure 3); the first is flagged weak and at the edge, the last two are clean bullseyes. The final peak is 22 m east of the burrow and 2.5 m lower. The paper reports “within about 20 m, uphill as expected”.

Ponui case 1. The four flights converge 195, 158, 42, 23 m (Figure 4); the last peak is 23 m north-north-east of the bush and 11 m lower, on a steep hillside. The paper reports “about 30 m”. The first two flights, where the bird may still have been elsewhere, put the peak 18 m *above* the final position; the two that found it put it below.



(a) Tag 40: the tag lies beyond the last transect, so the peak sits on it. (b) Tag 41: heard only near the noise floor; flagged weak.

Figure 2: Two behaviours the report flags, same flight as Figure 1.

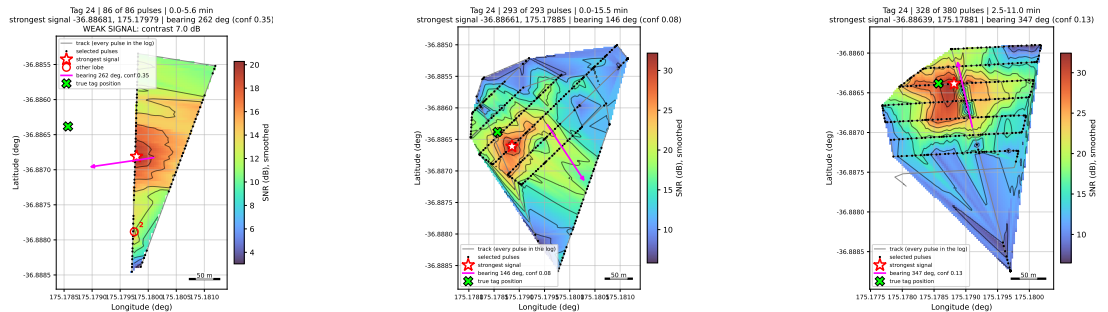


Figure 3: Ponui case 2, the three flights of 23 March 2025 on tag 24, converging 119, 36, 22 m on the burrow. The long wedge in the third panel is the convex hull stretched over the never-flown area between the grid and the transit leg; the spatial box is the control that cuts it.

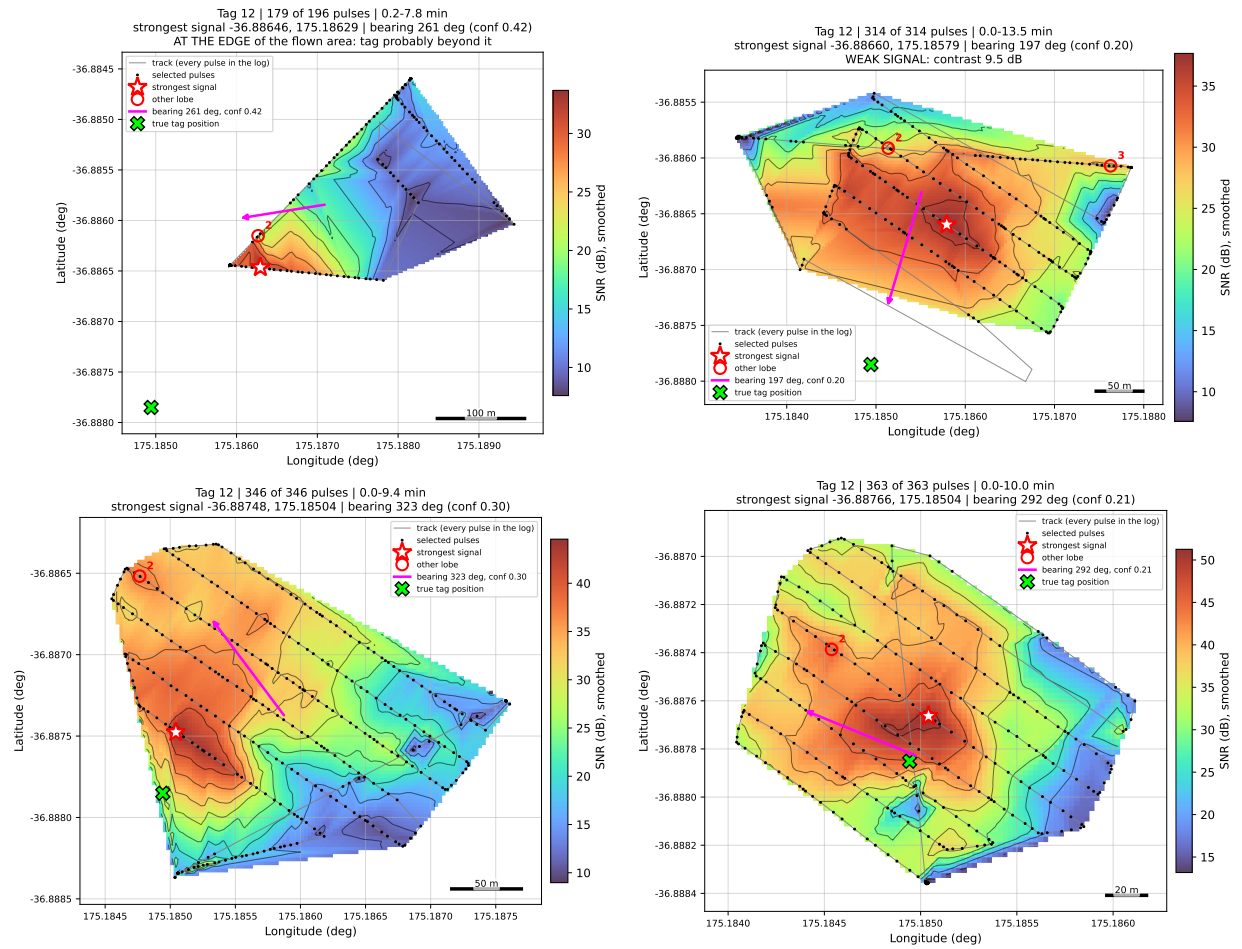


Figure 4: Ponui case 1, the four flights of 20 March 2025 on tag 12, converging 195, 158, 42, 23 m on the bush. The first flight's strongest signal is at the edge of its survey, as the paper describes.

4.3 Downhill?

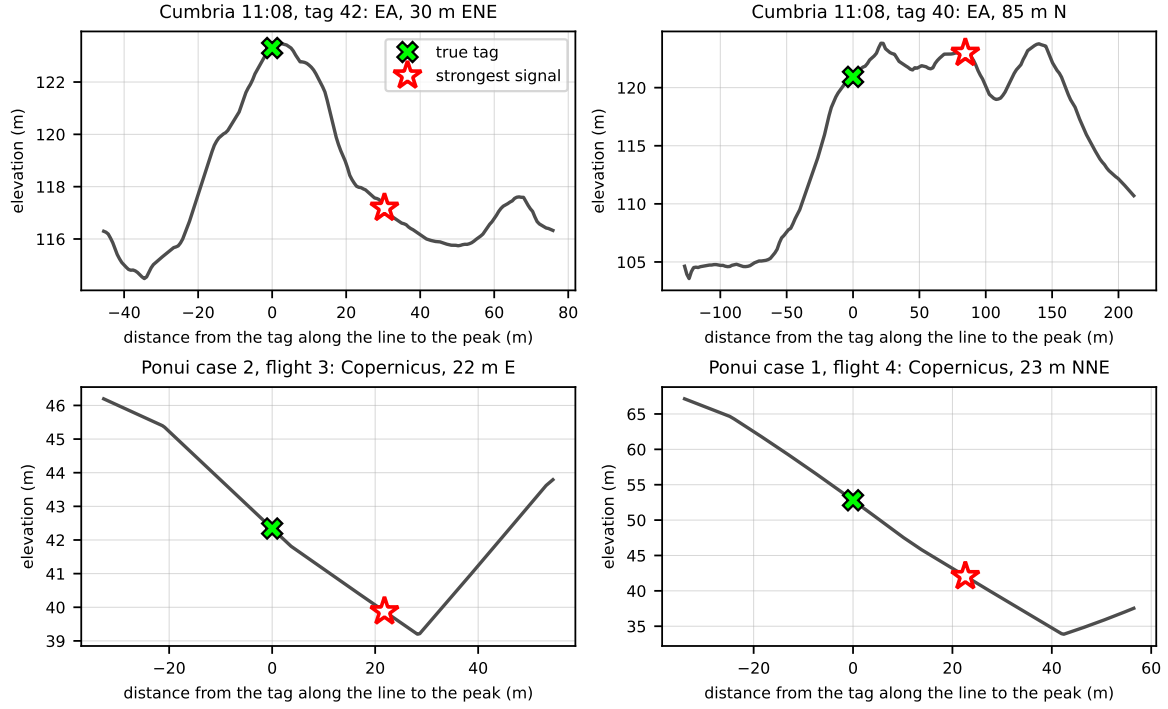


Figure 5: Terrain along the line from the true tag through the strongest signal, from the 1 m EA model (Cumbria) and the 30 m Copernicus model (Ponui).

On every well-sampled flight where the tag was inside the flown area the peak is lower than the tag: 3 m to 7 m on the three Cumbria tag 42 surveys, 2.5 m on the final Ponui case 2 flight, 11 m and 19 m on the last two case 1 flights. The one flight where the tag was outside the survey (tag 40) is the exception, and there the offset is the survey geometry. This is consistent with the paper’s account and its 20 m to 30 m residuals. It is also why the report never calls the position a tag position: a biologist reading “strongest signal, 30 m, downhill” will walk uphill from it, which is right.

4.4 Sensitivity to grid spacing and smoothing

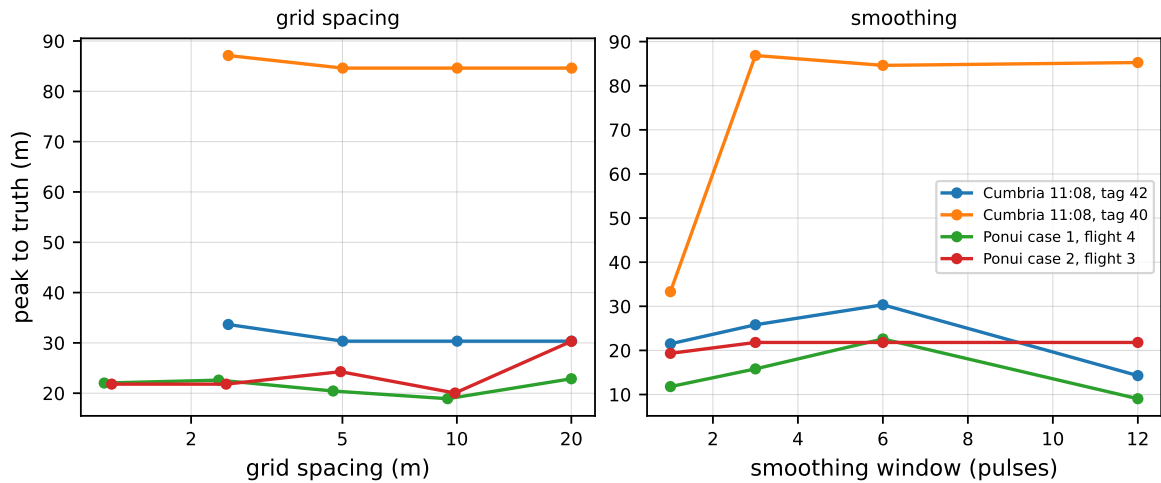


Figure 6: Distance from the strongest signal to the true tag as the grid spacing (left) and the smoothing window (right) are varied, four cases.

Case	Grid (m)	Peak shift (m)	Bearing shift	Offset (m)	Lobes
Cumbria 11:08, tag 42	2.5 rule $\times$ 0.5	4	-0	34	1
Cumbria 11:08, tag 42	5.0 rule $\times$ 1	0	+0	30	1
Cumbria 11:08, tag 42	10.0 rule $\times$ 2	0	+0	30	1
Cumbria 11:08, tag 42	20.0 rule $\times$ 4	0	+2	30	1
Cumbria 11:08, tag 40	2.5 rule $\times$ 0.5	3	+0	87	2
Cumbria 11:08, tag 40	5.0 rule $\times$ 1	0	+0	85	2
Cumbria 11:08, tag 40	10.0 rule $\times$ 2	0	+1	85	2
Cumbria 11:08, tag 40	20.0 rule $\times$ 4	0	+2	85	2
Ponui case 1, flight 4	1.2 rule $\times$ 0.5	2	+2	22	2
Ponui case 1, flight 4	2.4 rule $\times$ 1	0	+0	23	2
Ponui case 1, flight 4	4.7 rule $\times$ 2	2	-5	20	2
Ponui case 1, flight 4	9.5 rule $\times$ 4	5	-7	19	1
Ponui case 1, flight 4	20.0 fixed	10	+3	23	1
Ponui case 2, flight 3	1.2 rule $\times$ 0.5	0	-7	22	1
Ponui case 2, flight 3	2.5 rule $\times$ 1	0	+0	22	1
Ponui case 2, flight 3	4.9 rule $\times$ 2	2	+2	24	1
Ponui case 2, flight 3	9.9 rule $\times$ 4	6	+3	20	2
Ponui case 2, flight 3	20.0 fixed	22	-7	30	2

Table 3: Grid spacing from half to four times the rule, plus fixed values. Peak shift and bearing shift are relative to the rule’s own result.

Grid spacing barely matters, as the vertex argument predicts: over an eightfold range the peak moves by at most a cell or two, the offset to truth stays within 4 m on Cumbria and 8 m on Ponui, and the bearing moves by under 8° . Smoothing matters more, because it is what decides which pulse is the strongest: the offset ranges 14 m to 30 m on tag 42 and 33 m to 87 m on tag 40 as the window goes from 1 to 12 pulses. The bench default of 6 is kept; it belongs in the configuration menu, not on the screen.

4.5 Contours at small scale

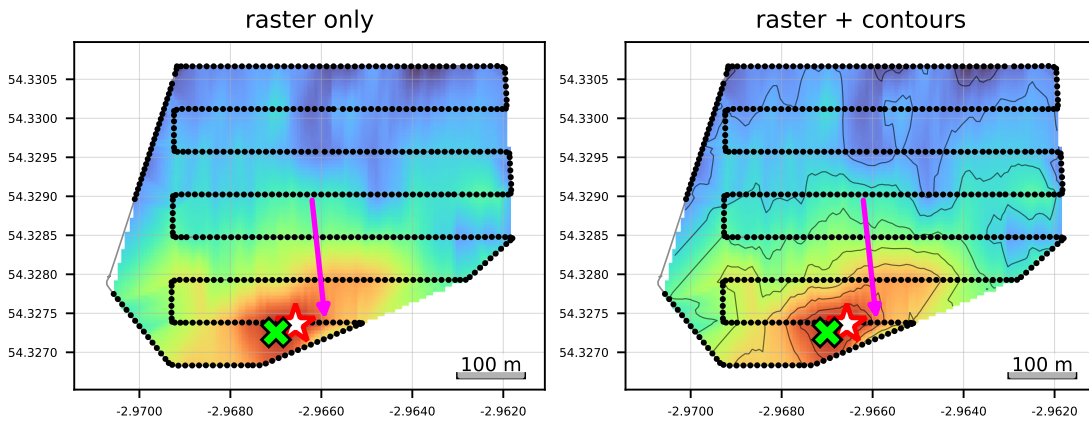


Figure 7: The tag 42 surface at roughly the size it would have on a 7 inch screen, without and with contour lines.

At this size the raster carries the message on its own; the contours add the shape of the peak and the direction of the slope, which is what the bearing is trying to say, and they do not obscure anything. They are cheap and switchable. Whether they survive sunlight on the actual Herelink is still a field question, but nothing here argues against carrying them.

4.6 Two lobes

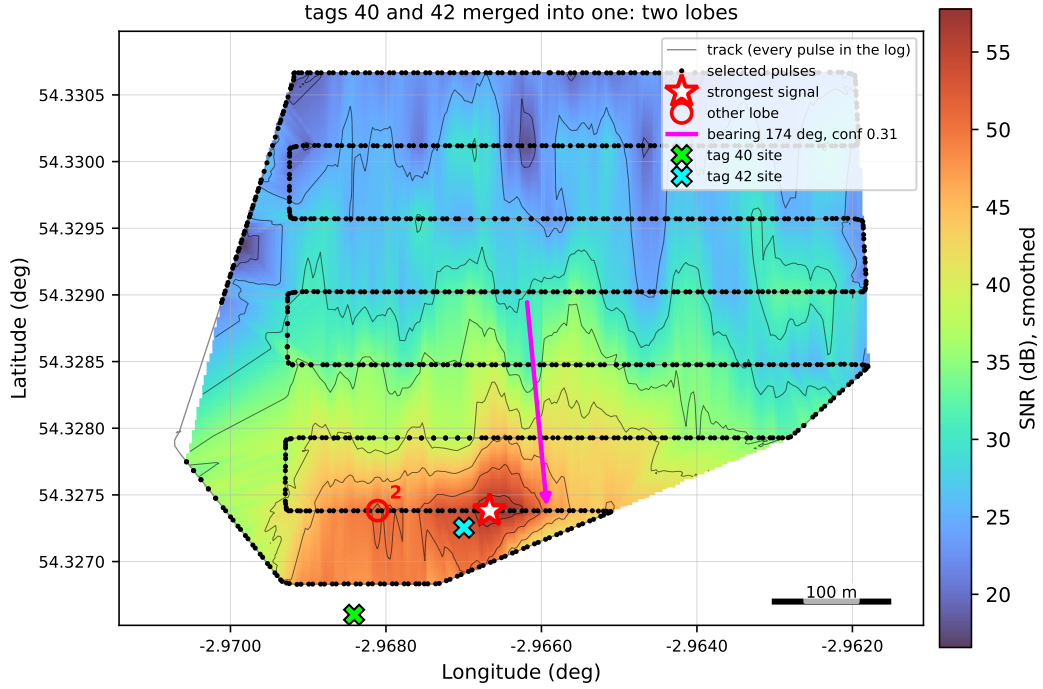


Figure 8: Cumbria tags 40 and 42, which sit 100 m apart, fed to the review as if they were one tag: what a log with two transmitters on one channel would look like. The report gives the strongest signal at the tag 42 site and a second lobe 94 m west at the tag 40 site, 6.5 dB down, out of 163 local maxima on the surface.

Case	Contrast	3 / 1	6 / 1	6 / 2	10 / 1	10 / 2	10 / 3	15 / 3
Cumbria 11:08, tag 42	31 dB	1	1	1	1	1	1	1
Cumbria 11:08, tag 43 (weak)	5 dB	1	1	1	1	1	1	1
Cumbria 11:08, tag 40	25 dB	1	2	2	2	2	2	2
Cumbria 11:08, tag 41 (weak)	9 dB	8	25	11	25	11	3	7
Cumbria 09:35, tag 42	14 dB	1	3	2	5	2	2	3
Cumbria 09:35, tag 40 (weak)	8 dB	1	5	3	6	4	2	3
Cumbria 13:05, tag 42	27 dB	1	1	1	1	1	1	1
Cumbria 13:05, tag 40	23 dB	1	2	2	2	2	2	4
Cumbria 14:12, tag 42	21 dB	1	1	1	1	1	1	2
Cumbria 14:12, tag 40	19 dB	2	2	1	2	1	1	1
Ponui case 1, flight 1	21 dB	1	2	2	2	2	2	3
Ponui case 1, flight 2 (weak)	9 dB	2	3	1	6	4	2	2
Ponui case 1, flight 3	16 dB	1	1	1	2	2	2	4
Ponui case 1, flight 4	15 dB	1	1	1	2	2	1	1
Ponui case 2, flight 1 (weak)	7 dB	1	3	2	3	2	1	1
Ponui case 2, flight 2	17 dB	1	2	1	2	1	1	1
Ponui case 2, flight 3	18 dB	1	1	1	1	1	1	1
Cumbria 11:08, tags 40+42 merged	28 dB	1	1	1	4	2	1	1

Table 4: Number of lobes each case reports as the window W / prominence floor P (dB) vary, uncapped. The defaults are 10/2.

5 Answers to the specification's questions

1. **What grid spacing rule works across a tight rotation and a wide lawnmower?** Equation (3): half the median pulse spacing, held between 60 and 300 cells across the longer side. It chose 2.2 m to 7.4 m on every survey on disk and fell back to the extent bound only on a 67 m lawnmower and on a log with duplicated positions. The position reported does not depend on it in any case.
2. **Do contours read at small scale, or is the raster enough?** The raster is enough to find the peak; the contours add shape and cost nothing. Keep them, switchable.
3. **What does the peak finder do with two lobes?** Reports both, ranked, with the second's distance, direction, depth below the first and prominence, using topographic prominence to tell a lobe from a wrinkle. Thresholds 10 dB window, 2 dB prominence, three lobes at most, from Table 4.
4. **How far is the reported peak from truth, and is it downhill?** 22 m to 41 m on well-sampled Cumbria flights, 22 m on the final Ponui case 2 flight, 23 m on the final case 1 flight; lower than the tag in every one of those, by 2.5 m to 19 m.

6 For the TagTracker delivery

- The analysis core is six functions with no GUI dependency and runs in under 0.2 s per flight on a laptop at 6 k to 80 k cells; the Herelink will be slower but the budget is generous.
- **Show the real track.** Here the track is the positions of every pulse in the log, because that is all the CSV holds. Inside TagTracker the vehicle position is free.
- Keep the caveat, the weak flag and the edge flag on the screen; they were right on every flight in the validation.
- The spatial box matters more than it looked. The convex hull stretches the surface over never-flown wedges wherever there is a transit leg (Figure 3); a box is the one-gesture fix.
- Show the bearing with its confidence, and expect it to be low on a bullseye.

7 Caveats and open items

- The 2023 logs write timestamps to six significant figures, so a whole flight shares one or two values; the time window and takeoff trim are meaningless on them.
- The confirmed-only filter cannot be tested on any log on disk.
- Copernicus is a surface model; on forested Ponui the terrain profiles include canopy.
- Prominence is implemented in a Python loop; the C++ port is straightforward.
- MATLAB was never run. All verification is Python against real logs and synthetic surfaces with known answers; `test_fieldreview.py` and `test_core.py` both pass.

A Command line

```
python fieldreview.py LOG [--tag ID] [--time START END] [--no-trim] [--alt LO HI]
                        [--box LAT0 LAT1 LON0 LON1] [--all] [--smooth N] [--grid M]
                        [--levels N] [--within-db DB] [--min-prominence DB]
                        [--max-peaks N] [--min-separation M] [--truth LAT LON]
                        [--png FILE] [--json FILE] [--show]
```

`-time` is minutes from the first pulse of the log. `-json` writes everything in the report plus the peaks, contours in lat/lon and the bearing, for another program to consume.

B Regenerating this document

```
cd DOCS
make fieldreview-figures      # runs ../python/validate_fieldreview.py; needs the logs
make fieldreview-prototype.pdf
```

The terrain checks download two rasters on first use into `python/data/dem/` (gitignored): the EA box for Cumbria and the Copernicus tile for Ponui. Pass `-no-dem` to `validate_fieldreview.py` to skip them.